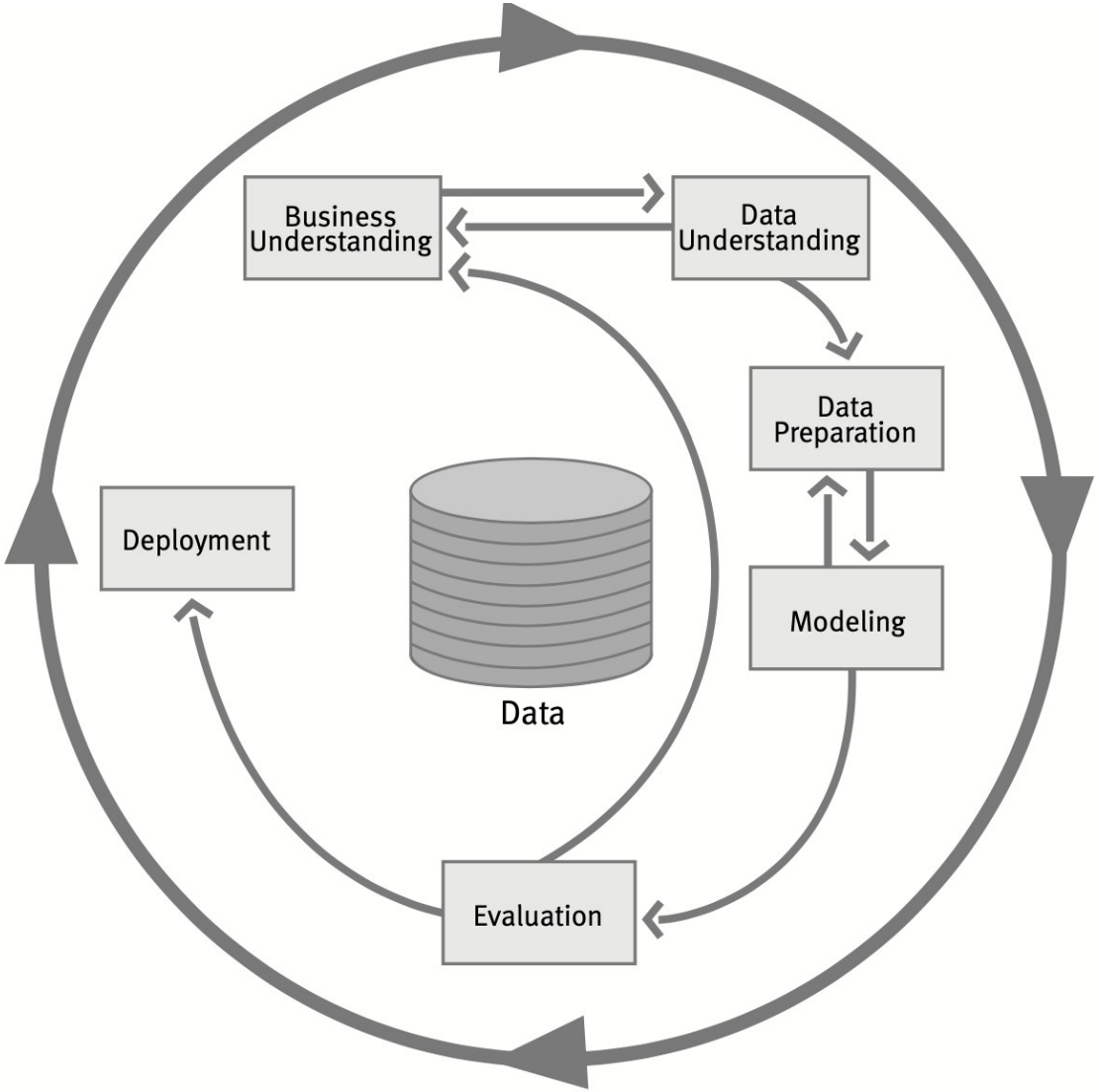
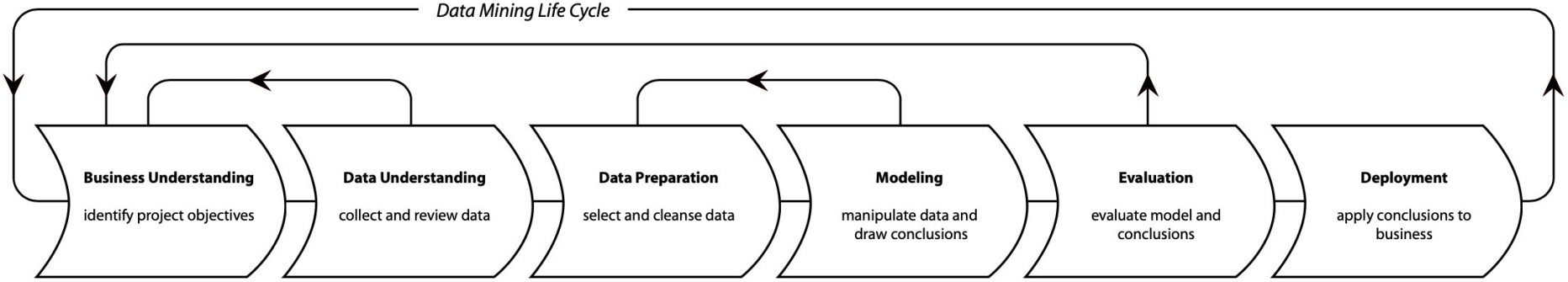
Business Analytics Framework

Problem > Algorithm


Business Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment
Data

Phases



Determine Business Objectives
Background
Business Objectives
Business Success Criteria
(Log and Report Process)

Assess Situation
Inventory of Resources, Requirements, Assumptions, and Constraints
Risks and Contingencies
Terminology
Costs and Benefits
(Log and Report Process)

Determine Data Mining Goals
Data Mining Goals
Data Mining Success Criteria
(Log and Report Process)

Produce Project Plan
Project Plan
Initial Assessment of Tools and Techniques
(Log and Report Process)

Collect Initial Data
Initial Data Collection Report
(Log and Report Process)

Describe Data
Data Description Report
(Log and Report Process)

Explore Data
Data Exploration Report
(Log and Report Process)

Verify Data Quality
Data Quality Report
(Log and Report Process)

Data Set
Data Set Description
(Log and Report Process)

Select Data
Rationale for Inclusion/ Exclusion
(Log and Report Process)

Clean Data
Data Cleaning Report
(Log and Report Process)

Construct Data
Derived Attributes
Generated Records
(Log and Report Process)

Integrate Data
Merged Data
(Log and Report Process)

Format Data
Reformatted Data
(Log and Report Process)

Select Modeling Technique
Modeling Technique
Modeling Assumptions
(Log and Report Process)

Generate Test Design
Test Design
(Log and Report Process)

Build Model Parameter Settings
Models
Model Description
(Log and Report Process)

Assess Model
Model Assessment
Revised Parameter
(Log and Report Process)

Evaluate Results
Align Assessment of Data Mining Results with Business Success Criteria
(Log and Report Process)

Approved Models
Review Process
Review of Process
(Log and Report Process)

Determine Next Steps
List of Possible Actions
Decision
(Log and Report Process)

Plan Deployment
Deployment Plan
(Log and Report Process)

Plan Monitoring and Maintenance
Monitoring and Maintenance Plan
(Log and Report Process)

Produce Final Report
Final Report
Final Presentation
(Log and Report Process)

Review Project
Experience Documentation
(Log and Report Process)

a visual guide to CRISP-DM methodology

Generic Tasks
Specialized Tasks
(Process Instances)

SOURCE CRISP-DM 1.0
http://www.crisp-dm.org/download.htm
DESIGN Nicole Leaper
http://www.nicoleleaper.com

BY NC ND

scikit-learn.org/stable/

Install User Guide API Examples More

Go

scikit-learn

Machine Learning in Python

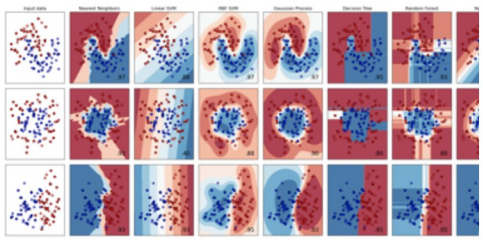
Getting Started | Release Highlights for 1.0 | GitHub

- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.
Algorithms: SVM, nearest neighbors, random forest, and more...

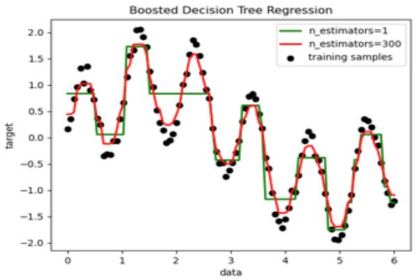


Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.
Algorithms: SVR, nearest neighbors, random forest, and more...

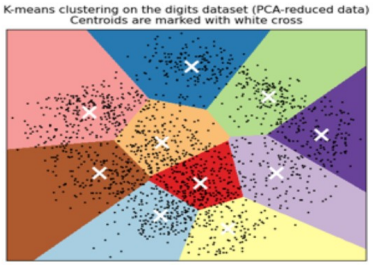


Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes
Algorithms: k-Means, spectral clustering, mean-shift, and more...



Examples

User Guide
1. Supervised learning
2. Unsupervised learning
3. Model selection and evaluation
3.1. Cross-validation: evaluating estimator performance
3.2. Tuning the hyper-parameters of an estimator
3.3. Metrics and scoring: quantifying the quality of predictions
3.4. Validation curves: plotting scores to evaluate models
4. Inspection
5. Visualizations
6. Dataset transformations
7. Dataset loading utilities
8. Computing with scikit-learn
9. Model persistence
10. Common pitfalls and recommended practices

3. Model selection and evaluation

3.1. Cross-validation: evaluating estimator performance

- 3.1.1. Computing cross-validated metrics
- 3.1.2. Cross validation iterators
- 3.1.3. A note on shuffling
- 3.1.4. Cross validation and model selection
- 3.1.5. Permutation test score

3.2. Tuning the hyper-parameters of an estimator

- 3.2.1. Exhaustive Grid Search
- 3.2.2. Randomized Parameter Optimization
- 3.2.3. Searching for optimal parameters with successive halving
- 3.2.4. Tips for parameter search
- 3.2.5. Alternatives to brute force parameter search

3.3. Metrics and scoring: quantifying the quality of predictions

- 3.3.1. The `scoring` parameter: defining model evaluation rules
- 3.3.2. Classification metrics
- 3.3.3. Multilabel ranking metrics
- 3.3.4. Regression metrics
- 3.3.5. Clustering metrics
- 3.3.6. Dummy estimators

3.4. Validation curves: plotting scores to evaluate models

- 3.4.1. Validation curve
- 3.4.2. Learning curve

„Machine Learning Model Evaluation”

e.g.
https://docs.aws.amazon.com/machine-learning/latest/dg/evaluating_models.html

Evaluating ML Models

PDF

You should always *evaluate a model* to determine if it will do a good job of predicting the target on new and future data. Because future instances have unknown target values, you need to check the accuracy metric of the ML model on data for which you already know the target answer, and use this assessment as a proxy for predictive accuracy on future data.

To properly evaluate a model, you hold out a sample of data that has been labeled with the target (ground truth) from the training datasource. Evaluating the predictive accuracy of an ML model with the same data that was used for training is not useful, because it rewards models that can "remember" the training data, as opposed to generalizing from it. Once you have finished training the ML model, you send the model the held-out observations for which you know the target values. You then compare the predictions returned by the ML model against the known target value. Finally, you compute a summary metric that tells you how well the predicted and true values match.

In Amazon ML, you evaluate an ML model by *creating an evaluation*. To create an evaluation for an ML model, you need an ML model that you want to evaluate, and you need labeled data that was not used for training. First, create a datasource for evaluation by creating an Amazon ML datasource with the held-out data. The data used in the evaluation must have the same schema as the data used in training and include actual values for the target variable.

If all your data is in a single file or directory, you can use the Amazon ML console to split the data. The default path in the Create ML model wizard splits the input datasource and uses the first 70% for a training datasource and the remaining 30% for an evaluation datasource. You can also customize the split ratio by using the **Custom** option in the Create ML model wizard, where you can choose to select a random 70% sample for training and use the remaining 30% for evaluation. To further specify custom split ratios, use the data rearrangement string in the Create Datasource API. Once you have an evaluation datasource and an ML model, you can create an evaluation and review the results of the evaluation.

Topics

- ML Model Insights
- Binary Model Insights
- Multiclass Model Insights
- Regression Model Insights
- Preventing Overfitting
- Cross-Validation
- Evaluation Alerts

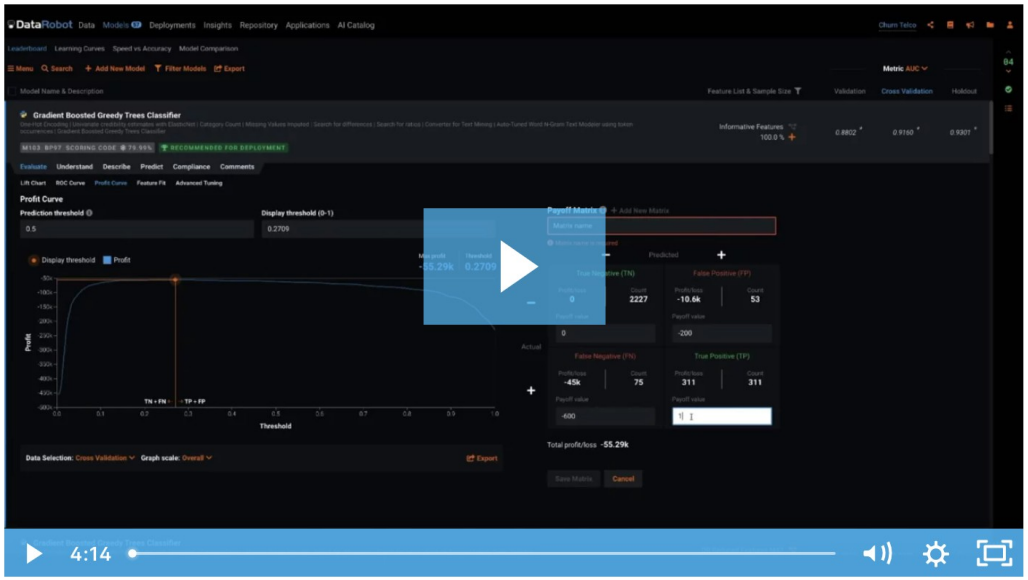
ML quality > Business Impact

Profit Curves

https://community.datarobot.com/t5/knowledge-base/profit-curve/ta-p/7389

Profit Curve

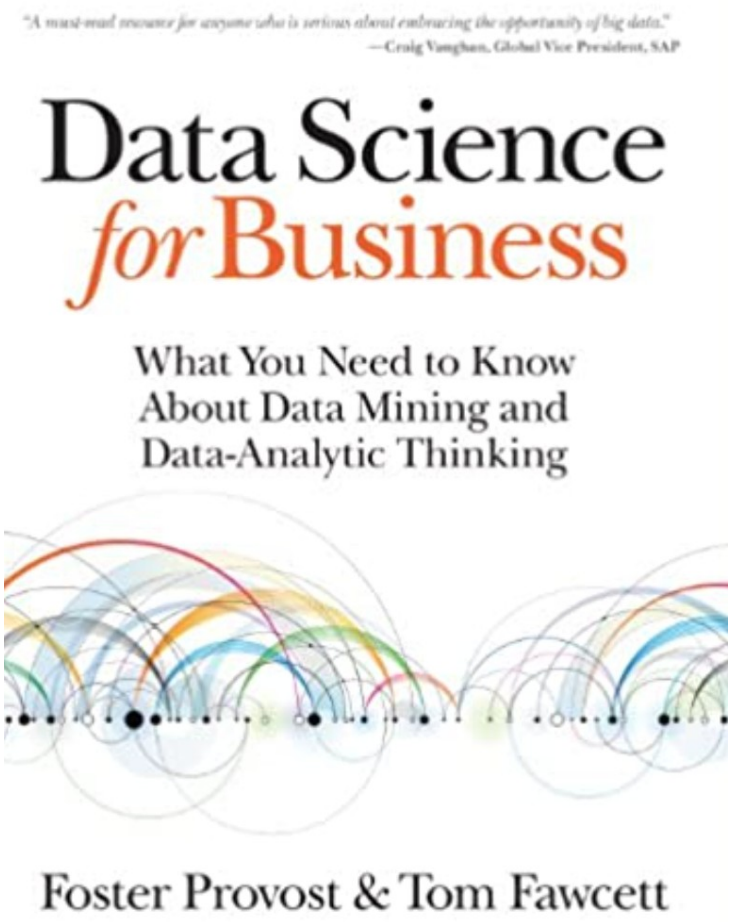
This article explains how to use the Profit Curve in binary classification projects to determine the business impact of a selected model. The Profit Curve allows assessment of a model based on your supplied costs or benefits and helps to visualize how those profits change with different inputs. The matrix informing the Profit Curve is called the profit Payoff Matrix, which is built based on the same four quadrants as those found in the confusion matrix: false positive, false negative, true positive, and true negative. Unlike the confusion matrix that lives in the **ROC Curve** tab, the profit Payoff Matrix lives in the **Profit Curve** tab (from the **Evaluate** tab for a selected model).



ML quality > Business Impact

Profit Curves, and other very useful insights

https://www.amazon.com/Data-Science-Business-Data-Analytic-Thinking-ebook/dp/B00E6EQ3X4/ref=sr_1_1?crid=163UTLIDEYJVG&dchild=1&keywords=data+science+for+business+provost+%26+fawcett&qid=1634464425&sprefix=provost+faw%2Caps%2C356&sr=8-1



Business Analytics Framework

Problem

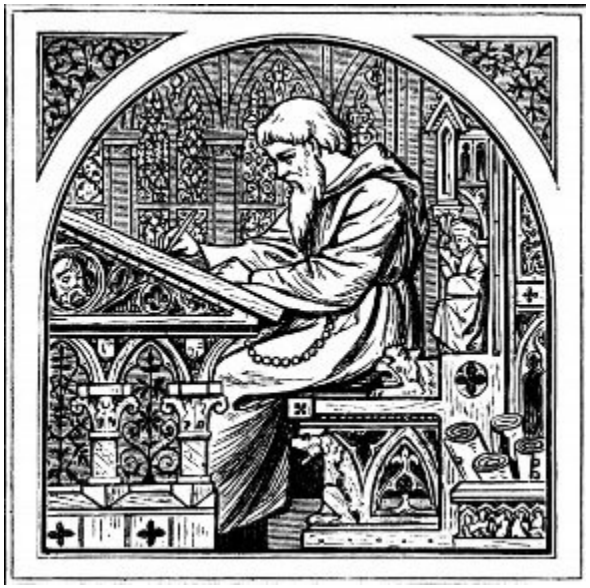
Predict					Analyze
Data **rich**				Data **poor**	Descriptive
Numeric		Classification		A/B Testing	Geospatial
Continuous	Time-based	2 binary	>2 non-binary		Segmentation
					Aggregation
Linear regression Decision trees Random forest Boosted models	Time-series forecasting	Logistic regression Decision trees	Random forests Boosted models		
Regression		*Classification*		*Experiment*	*Data Analytics*

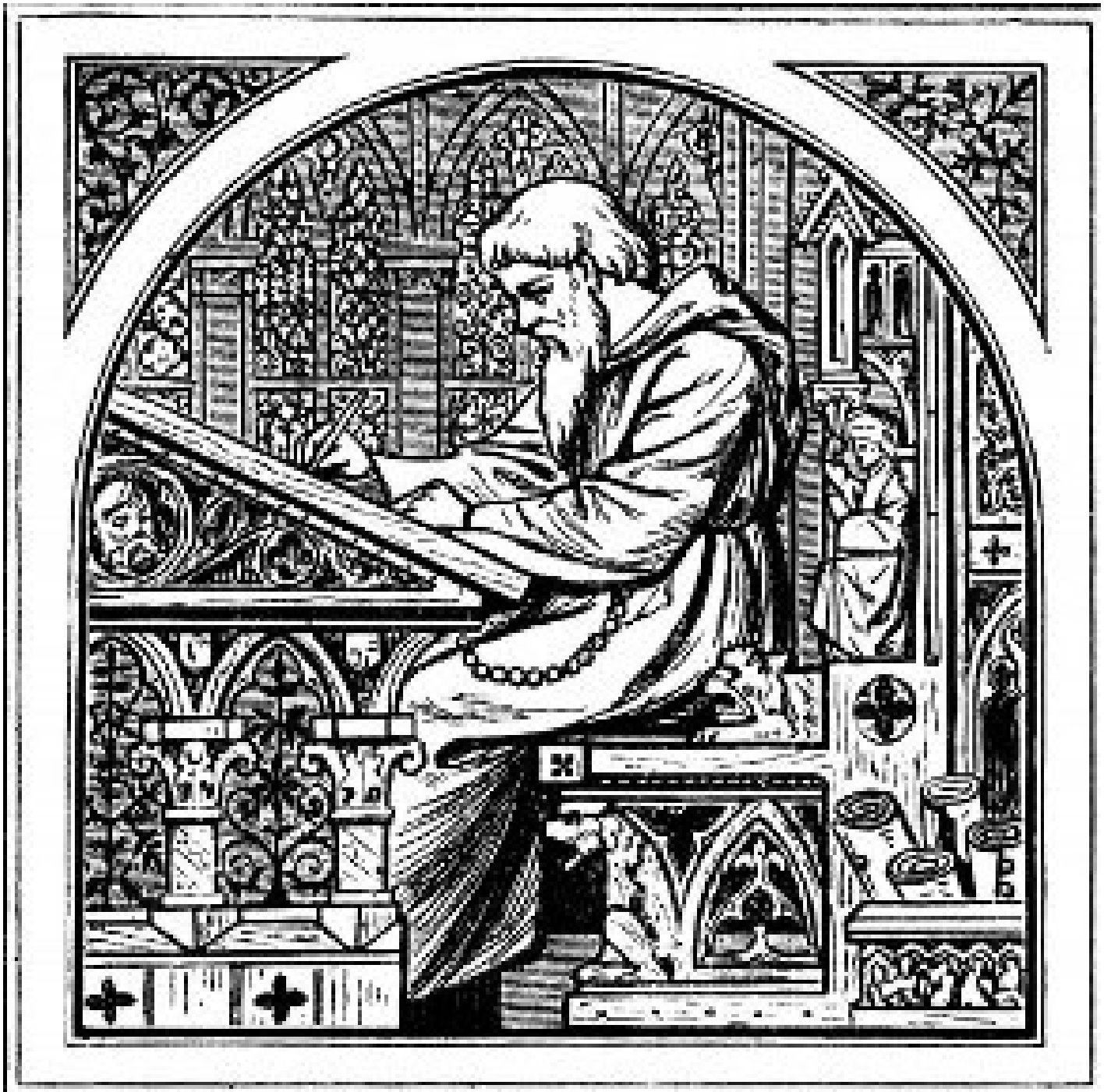
inspiried by: https://eu.udacity.com/course/problem-solving-with-advanced-analytics--ud976

What do you want?

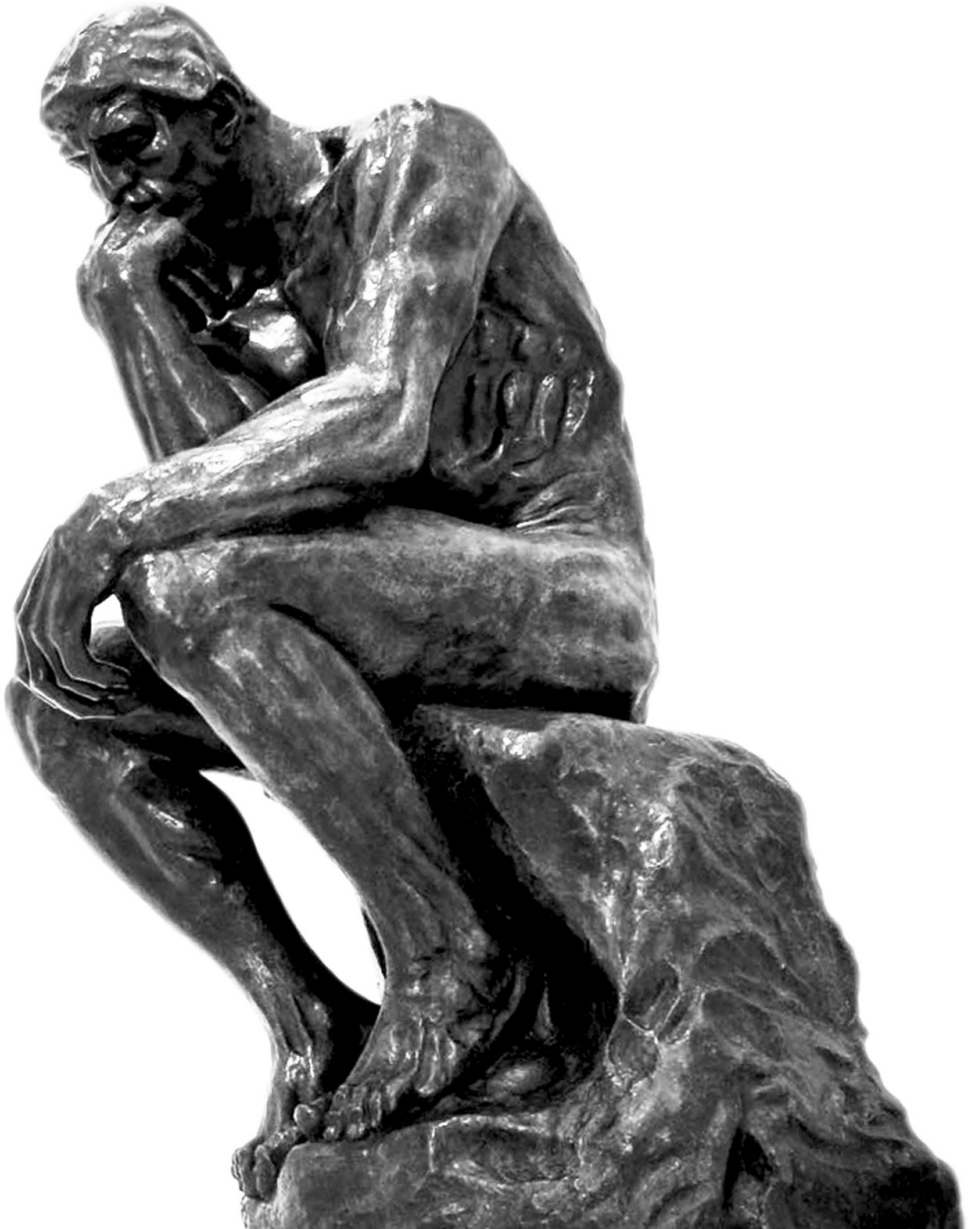
Predict something?

Or just analyze data?





What’s the main purpose?



Geospatial analysis?

Group customers by location
(np. postal code, county, region)

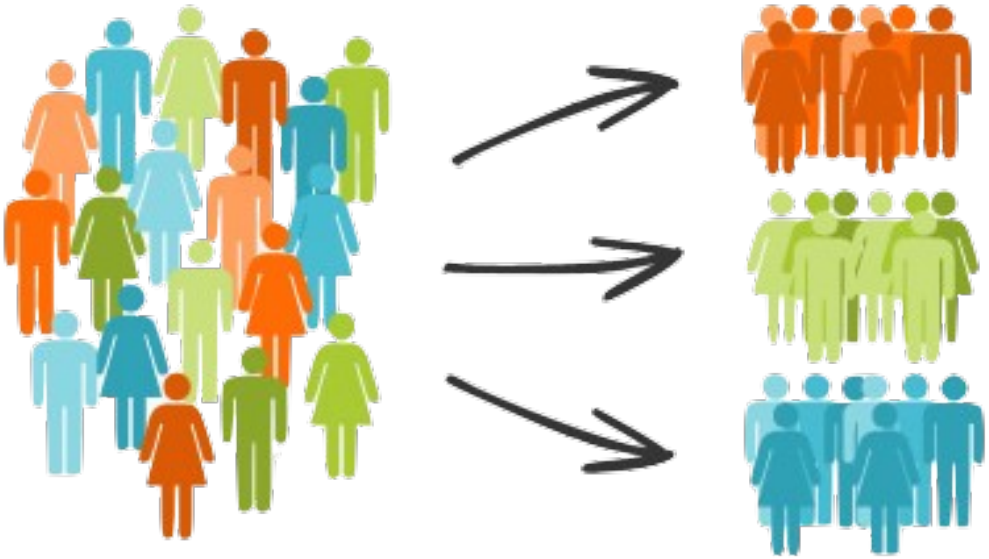
Compute distance between shops

Optimize shops location

Segmentation?

1. Identify objects categories
2. Assign objects to these categories

Customer who bought this also bought…
Shops selling goods to customers of a given category



Aggregation?

Compute diferent quantities with respect to some dimensions

… by …

Sales by salesman
Sales by number of citizens
Sales by month by region

Just describe data ?

Simple descriptive analytics

Average
Median
Standard deviation
Distributions (Gauss, etc.)

Grocery stores

BlaBla Mart wants to build another store in the area where they will have the most business, but they want to make sure that it is not within six miles of another location.

- prediction
- ✓ geo-location
- segmentation
- aggregation
- descriptive analytics

E-commerce site

Somenicestuff.com received complaints that their web pages are loading slowly.

They want to understand what web traffic levels to expect in the future in order to ensure they have enough server capacity to handle the traffic.

- ✓ prediction
- geo-location
- segmentation
- aggregation
- descriptive analytics

Shoe market

Managers from someniceshoooooooes.com want to know how much they ship to each zip code, state, or region

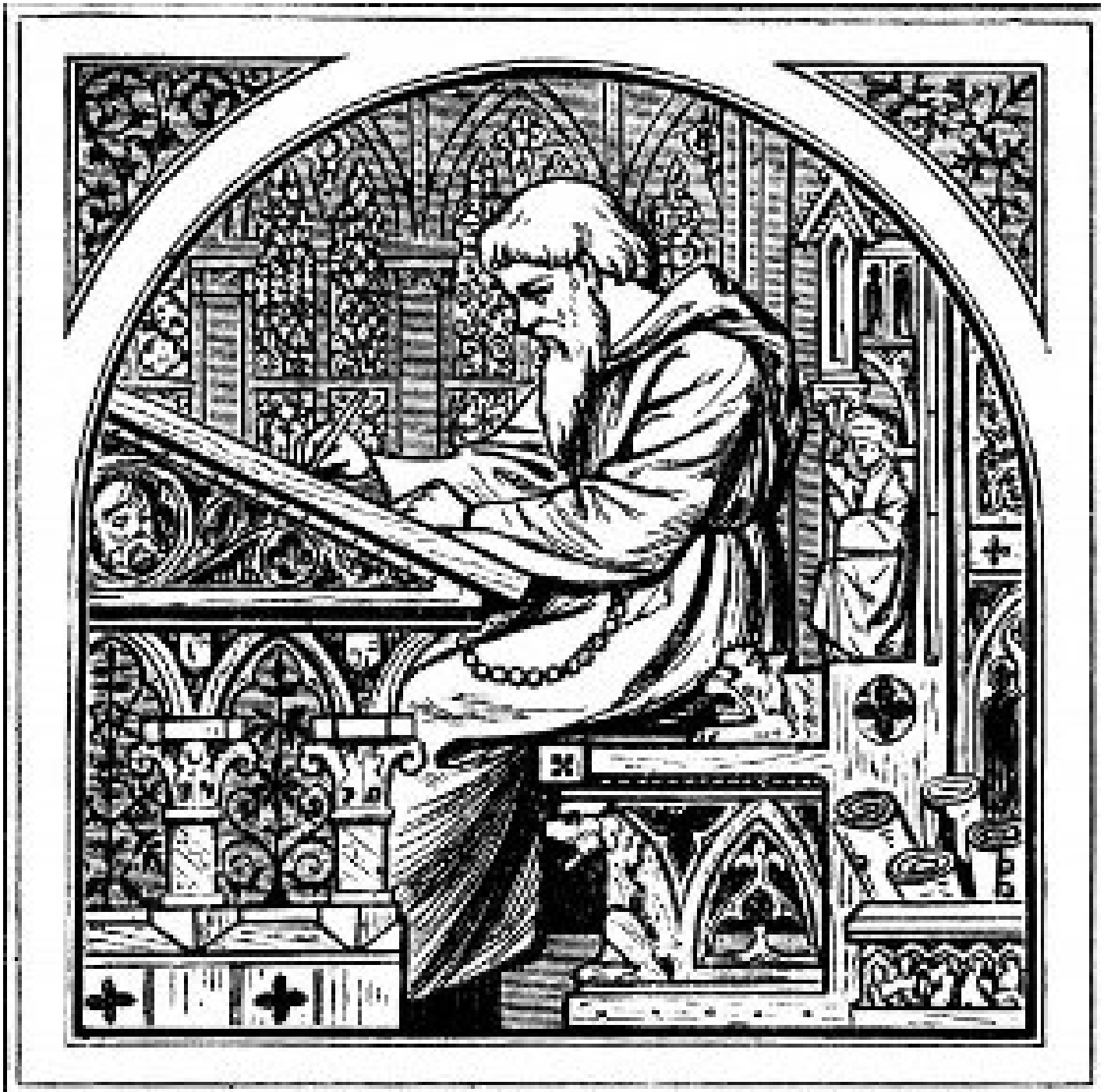
- prediction
- geo-location
- segmentation
- ✓ aggregation
- descriptive analytics

Clothes company

Beauty Pinky T-shirts wants to market to their top 3 customer segments.

They need to create the customer segments, calculate how many customers fall into each one, and then determine which segment has the most customers

- prediction
- geo-location
- ✓ segmentation
- ✓ aggregation
- descriptive analytics



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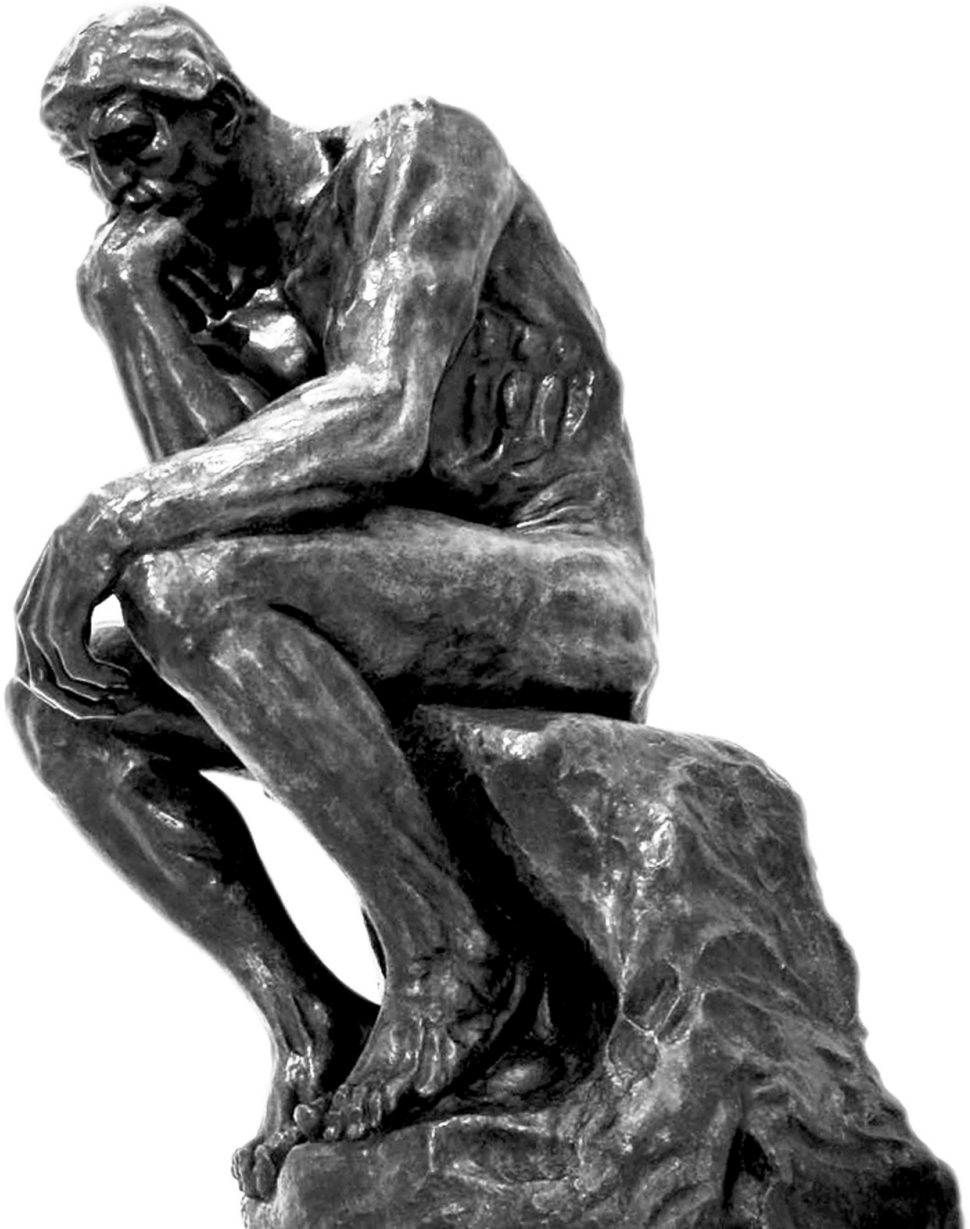
How many data do we have?

A lot?

Or close to nothing?



Data poor

We must experiment…

A/B testing

You introduce a new product to the market and want to estimate it's sales dynamics

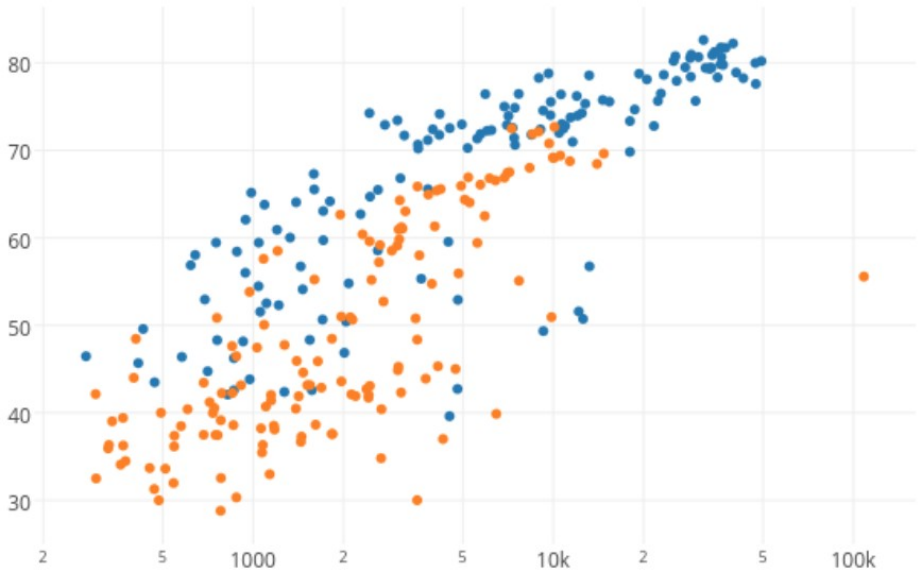
You want to estimate the results of the new TV marketing campaign having no previous experience with that media

Data rich

Ice-cream vendor: how much ice-cream should I order for next weekend (sales differ from week to week)?

Bank: how many customers who applied for a loan in fact will finalize the deal?

Clothes store: how many of the customer I sent a promotion coupon to will realize it?



What is a type of a variable I want to predict?

Number?

How much? How many?

How many bicycles should we produce next quarter to fulfill a demand?

How many pizzas we will probably sell in our new location (where we know the demography)?

…regression…

Category?

Is…? What? Which?

Will this new customer pay this loan?

Is this a cancer?

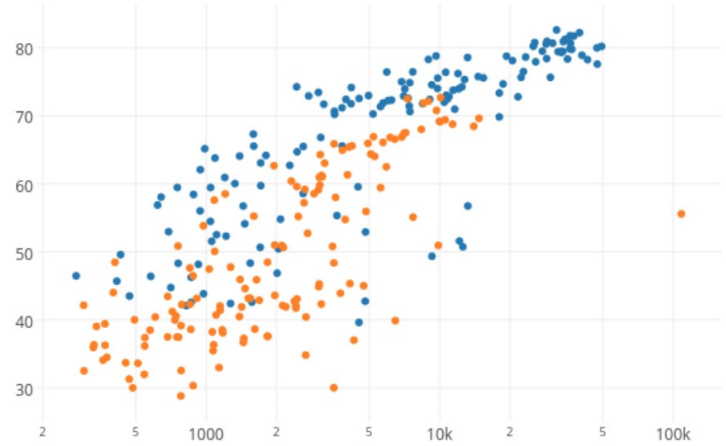
… classification…

1500 100 900

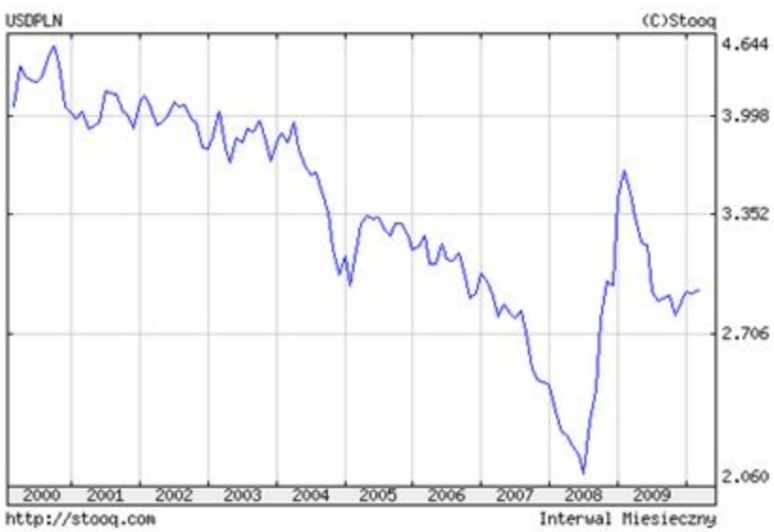
Regression

What is a type of a variable we want to predict?

Continuous
Continuous models



Time series
Time-series analysis and forecasting



Classification

How many categories do we have?

2: binary classification

Will this customer pay her loan?
Will this tire break in next 100 days?

More than 2: multiclass classification

Is this shop small, medium or large?

What's a breed of this dog?

Problem

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Bank

Bank is considering changing it's branches opening times

Before the implementation it wants to estimate it's influence on operational efficiency.

continuous, numeric

time-based

binary classification

non-binary classification

A/B testing

descriptive analysis

Shoe maker

Shoe maker wants to identify customer, who will apply discount coupons they receive during promotion.

- continuous, numeric
- time-based
- ✓ binary classification
- non-binary classification
- A/B testing
- descriptive analysis

Shoe maker

The same shoe maker wants to predict the sales growth generated by the marketing folders sent to his customers.

- ✓ continuous, numeric
- time-based
- binary classification
- non-binary classification
- A/B testing
- descriptive analysis

Zoo shop

Zoo shop wants to predict the sales from it's new store

- ✓ continuous, numeric
- time-based
- binary classification
- non-binary classification
- A/B testing
- descriptive analysis

Bank

Bank wants to identify who will pay a loan according to it's terms (no delay)

- continuous, numeric
- time-based
- ✓ binary classification
- non-binary classification
- A/B testing
- descriptive analysis

Coffee network

Coffee network wants to estimate the revenue generated by a new TV marketing campaign

- continuous, numeric
- time-based
- binary classification
- non-binary classification
- ✓ A/B testing
- descriptive analysis

Restaurant network

Restaurant network wants to calculate sales and profits generated by it's restaurant in a previous year

- continuous, numeric
- time-based
- binary classification
- non-binary classification
- A/B testing
- ✓ descriptive analysis

On-line game vendor

On-line game vendor wants to estimate the demand for games in next months to adapt it's servers and network capacity

- continuous, numeric
- ✓ time-based
- binary classification
- non-binary classification
- A/B testing
- descriptive analysis

Fiance

You want to buy a diamond ring to Your fiancee. Based on a database you want to estimate the best price for a diamond with given parameters

- ✓ continuous, numeric
- time-based
- binary classification
- non-binary classification
- A/B testing
- descriptive analysis

University

University wants to identify students which are at risk of drop-out

- continuous, numeric
- time-based
- ✓ binary classification
- non-binary classification
- A/B testing
- descriptive analysis

Application developer

App developer wants to check if changing a graphic button to a link will increase a number of downloads

- continuous, numeric
- time-based
- binary classification
- non-binary classification
- ✓ A/B testing
- descriptive analysis

Hotel

Hotel wants to predict the number of reservations next year to plan it's infrastructure development

- continuous, numeric
- ✓ time-based
- binary classification
- non-binary classification
- A/B testing
- descriptive analysis

More questions to ask…

https://blogs.technet.microsoft.com/machinelearning/2015/09/01/which-algorithm-family-can-answer-my-question/

	Two-class classification	Multi-class classification	Regression	Anomaly detection	Unsupervised learning	Reinforcement learning
● first choice ○ may also work well						
Predictive Maintenance						
Should I replace this part now?	●		○			
When will this part fail?		○	●			
Will this tire fail in the next thousand miles?	●		○			
Is this unit behaving in an unusual way?	○			●		
What is the remaining useful life of this aircraft engine?			●			
Which vehicle needs servicing most urgently?		●	○			○
Is this pressure reading unusual?	○			●		
Are these voltages normal for this season and time of day?	○			●		
Is this internet message typical?	○			●		
Is this network activity part of an attack?	●					
Out of a thousand units, how many of this model of bearings will survive 10,000 hours of use?		○	●			

● **first choice** ○ **may also work well**

	Two-class classification	Multi-class classification	Regression	Anomaly detection	Unsupervised learning	Reinforcement learning
Marketing						
Will my customer leave me for a competitor?	●		○			
When will this customer make another purchase?		○	●			
Will this customer renew their subscription?	●	○				
Of all my customers, which 10% should receive an offer?	○		●			
Should this customer receive a promotional offer?	●		○			
Will this customer respond to this marketing campaign?	●					
Which offer should this customer receive?		●				○
Is the customer amenable to upselling?	●					
Will this customer tip for their taxi ride?	●					
Does the $5 coupon or the 25% off coupon result in more return customers?	●		○			
How likely is this person to make a purchase?	●		○			

● first choice ○ may also work well	Two-class classification	Multi-class classification	Regression	Anomaly detection	Unsupervised learning	Reinforcement learning
It this review positive or negative?	○		●			
Is the overall Twitter mood on my business positive or negative?	●		○			
How many new followers will I get next week?			●			
Which advertisement should be listed first for this reader?			○			●
Will this customer click on the top link?	●		○			
Is this combination of purchases very different from what this customer has made in the past?	○			●		
What will my fourth quarter sales be for the nation?			●			
What fraction of pulls on this slot machine result in payout?		○	●			
What other products is this customer likely to buy?		●	○			
Which other customers have similar preferences to this one?					●	
What is a natural way to divide this set of customers into groups?					●	
How highly rated will this wine be?		●	○			

● **first choice** ○ **may also work well**

	Two-class classification	Multi-class classification	Regression	Anomaly detection	Unsupervised learning	Reinforcement learning
Operational Efficiency						
What will the demand for this item (or service) be next month?		○	●			
How many bikes will be rented in the next hour?			●			
How much beer will be consumed at this event?			●			
What price should I set on this item?			●			○
Is this flight going to be on time?	●					
What fraction of today's flights will depart on time?		○	●			
How many employees should be scheduled to work on Black Friday?		○	●			
Is it time to order more of this product?	●					○

● **first choice** ○ **may also work well**

	Two-class classification	Multi-class classification	Regression	Anomaly detection	Unsupervised learning	Reinforcement learning
Internet of Things						
Has this patient's health suddenly taken a turn for the worse?	○			●		
Is your heart rate within your typical training range?	○			●		
What activity is the wearer of a fitness tracker engaged in?		●				
Should the robot vacuum clean the living room or continue to charge?	○					●
Do I move this obstacle or navigate around it?	○					●
Which aircraft is causing this radar signature?		●				
Who is the speaker in this recording?		●				
What will the temperature be next Tuesday?			●			
Will an earthquake occur in this city in the next year?	●					
Should the thermostat adjust the temperature higher, lower, or leave it where it is?		○				●
Should I continue driving at the same speed,		○				●